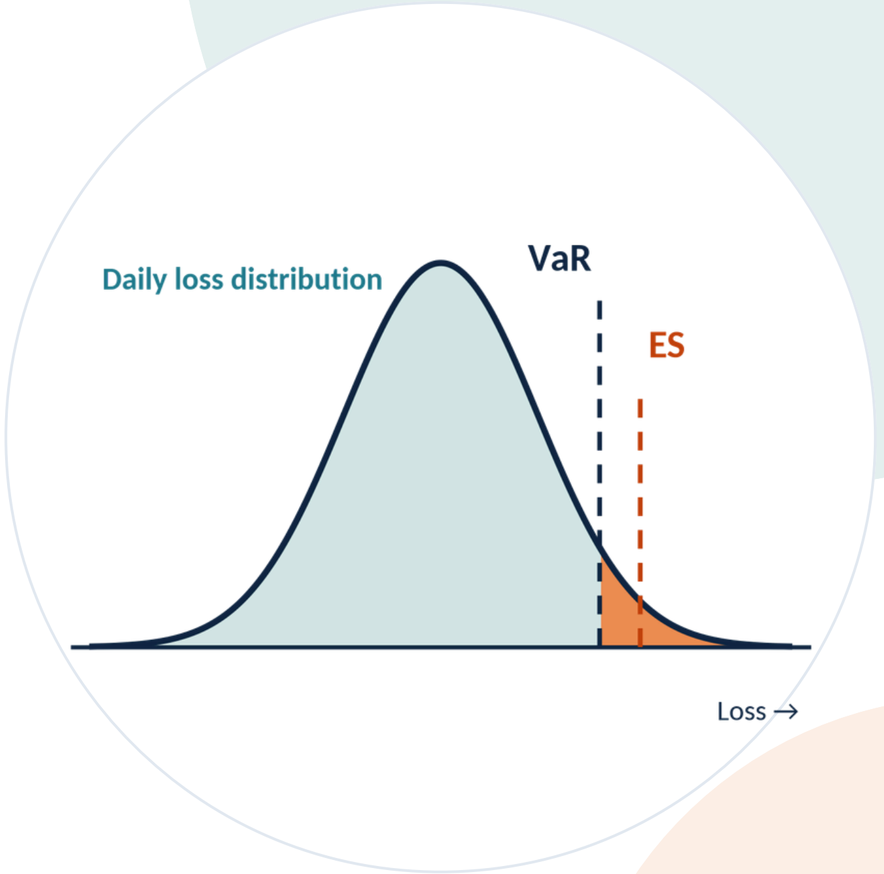
INTERNSHIP YEAR 4 | DEFENSE DAY, 23TH-24TH

Financial Risk Management of Food & Beverage and Retail Sales

A Quantitative Risk Model Using Value at Risk, Stress Testing and Store-Level Risk Scoring



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ACADEMIC YEAR
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1 Host organization: Newrest Cambodia

THE COMPANY

Newrest Cambodia

The study analyses point-of-sale records from its restaurants, cafés and retail shops, covering food, beverages and retail products.

21 outlets analysed

4 product categories

PRODUCT CATEGORIES IN THE SALES DATA

1 **Food**
FINFOOD

2 **Beverages**
FINBEVERAG

3 **Alcoholic drinks**
FINALCOHOL

4 **Retail goods**
RETAIL

2 Dataset: Sales.csv

2,000,000
sales lines in the raw file

21
stores

Jan – Dec 2022
1 January to 31 December

1,998,500
rows kept after validation

FIELDS IN EACH RECORD

- Store
- Ticket
- Date and time
- Product identifiers
- Category
- Quantity
- Gross sale
- Net sale
- Discount
- Payment method
- VAT
- Unit cost
- Unit price

DATA QUALITY

481,140
exact duplicate rows (24.06%), flagged but not deleted

1,500
rows with non-positive net sales (0.07%), excluded from the modelling sample

3 Objectives

GENERAL OBJECTIVE

Build and document a quantitative risk model for the food, beverage and retail sales of the organisation.

SPECIFIC OBJECTIVES

1. Validate and clean the transaction data and document every exclusion rule.
2. Build daily revenue, cost, profit and loss variables and a business-performance baseline.
3. Estimate Historical, Parametric and Monte Carlo VaR and Expected Shortfall (ES) on the same one-day loss definition, and compare them.
4. Stress test annual profit under sales declines and cost increases of up to 20%.
5. Measure concentration, discount, margin and demand risks and combine them in an entropy-weighted risk score for each store.
6. Translate the results into a risk register and management recommendations.

4. DETAIL CONCEPT

Financial Risk Management of Food & Beverage and Retail Sales

FORMULAS IN THIS DECK

$$\mathrm{VaR}_c = \mu_L + z_c\,\sigma_L$$

$$\mathrm{ES}_c = \frac{1}{N_c} \sum_{i:\ L_i \geq \mathrm{VaR}_c} L_i$$

$$\mathrm{HHI} = \sum_{i=1}^{n} s_i^2 \times 10^4$$

$$CRS_s = \sum_{j=1}^{5} w_j\, S_{sj}$$

5 Transaction-line features

Cost

$$\mathrm{Cost}_t = \mathrm{UnitCost}_t \times \mathrm{Quantity}_t$$

Profit

$$\mathrm{Profit}_t = \mathrm{NetSale}_t - \mathrm{Cost}_t$$

Margin rate

$$\mathrm{MarginRate}_t = \frac{\mathrm{Profit}_t}{\mathrm{NetSale}_t}$$

Discount rate

$$\mathrm{DiscountRate}_t = \frac{\mathrm{Discount}_t}{\mathrm{GrossSale}_t}$$

PARAMETER MEANINGS

t	Identifies a transaction line	$\mathrm{UnitCost}_t$	Cost of one unit on line t	$\mathrm{Quantity}_t$	Units sold on line t
$\mathrm{GrossSale}_t$	Revenue before discount	$\mathrm{Discount}_t$	Amount discounted on line t	$\mathrm{NetSale}_t$	Revenue collected after discount
Cost_t	Unit cost times quantity	Profit_t	Net sale minus total cost	$\mathrm{MarginRate}_t$	Profit divided by net sale
$\mathrm{DiscountRate}_t$	Discount divided by gross sale				

5.1 Daily profit and daily loss

Daily profit

$$\text{DailyProfit}_t = \sum_i \text{Profit}_{i,\,t}$$

Profit change

$$\Delta\text{Profit}_t = \text{DailyProfit}_t - \text{DailyProfit}_{t-1}$$

Daily loss

$$\text{Loss}_t = -\Delta\text{Profit}_t$$

PARAMETER MEANINGS

t	Identifies a day	i	A transaction occurring on day t	$\text{Profit}_{i,t}$	Profit of transaction i on day t
DailyProfit_t	Total profit of day t	ΔProfit_t	Change in profit from day t−1 to day t	Loss_t	Negative of the profit change; a fall is a positive loss

5.2 Sample mean and standard deviation

Sample mean

$$\bar{L} = \frac{1}{n} \sum_{t=1}^{n} L_t$$

Sample standard deviation

$$s = \sqrt{\frac{1}{n-1} \sum_{t=1}^{n} (L_t - \bar{L})^2}$$

PARAMETER MEANINGS

n	Number of daily-loss observations
L_t	Loss recorded for observation t
$\bar{L}$	Sample average of the daily losses
s	Sample standard deviation of the losses

5.2 Skewness and excess kurtosis

Skewness

$$G_1 = \frac{n}{(n-1)(n-2)} \sum_{t=1}^{n} \left(\frac{L_t - \bar{L}}{s} \right) 3$$

Excess kurtosis

$$EK = \frac{\frac{1}{n} \sum_{t=1}^{n} (L_t - \bar{L})^4}{\left[\frac{1}{n} \sum_{t=1}^{n} (L_t - \bar{L})^2 \right]^2} - 3$$

PARAMETER MEANINGS

n	Number of daily-loss observations	L_t	Loss recorded for observation t	$\bar{L}$	Sample average of the losses
s	Sample standard deviation	G_1	Sample skewness: asymmetry of the loss distribution	EK	Excess kurtosis = kurtosis – 3; zero means Normal tails

5.2 Jarque-Bera normality test

Jarque-Bera statistic

$$JB = \frac{n}{6}\left(S^2 + \frac{EK^2}{4}\right), \qquad JB \sim \chi^2(2)$$

p-value, 2 degrees of freedom

$$p = \exp\left(-\frac{JB}{2}\right)$$

PARAMETER MEANINGS

n	Number of observations in the sample	S	Sample skewness	EK	Sample excess kurtosis
JB	Jarque-Bera test statistic	$\chi^2(2)$	Chi-square law with 2 degrees of freedom under Normality	p	p-value: reject Normality if p < 0.05

5.3 Historical simulation VaR

Definition of VaR at level c

$$P(L > \mathrm{VaR}_c) = 1 - c$$

Empirical quantile

$$\mathrm{VaR}_{i,t}(\tau) = Q_{r^w_{i,t}}(\tau), \qquad \tau = 1 - c$$

Order statistic

$$Q(1-c) = x_{(\lceil n(1-c) \rceil)}$$

Loss VaR (sign reversed)

$$\mathrm{VaR}_c = -Q(1-c)$$

PARAMETER MEANINGS

c	Confidence level (95% or 99%)	$\tau = 1 - c$	Probability in the adverse tail	L	Daily loss; positive when profit falls
VaR_c	Loss threshold at confidence level c	r	Risk factor: the daily profit changes	w	Length of the observation window
$Q(\tau)$	Empirical quantile at probability τ	n	Number of observations used	$\lceil \cdot \rceil$	Ceiling: round up to the next integer
$x_{(k)}$	k-th smallest observation after sorting				

5.3 Expected Shortfall

Average loss beyond VaR

$$\mathrm{ES}_c = \frac{1}{N_c} \sum_{i\,:\,L_i \geq \mathrm{VaR}_c} L_i$$

PARAMETER MEANINGS

c	Confidence level of the VaR threshold	ES_c	Average loss at or beyond the VaR threshold	VaR_c	Loss threshold at confidence level c
N_c	Number of days with a loss of at least VaR	L_i	Loss on observed day i	i	Index of the days included in the tail

5.4 Parametric (Normal) VaR and ES

Normal loss model

$$L \sim N(\mu_L, \sigma_L^2)$$

Parametric VaR

$$\mathrm{VaR}_c = \mu_L + z_c \sigma_L$$

Parametric ES

$$\mathrm{ES}_c = \mu_L + \sigma_L \frac{\phi(z_c)}{1-c}, \qquad z_c = \Phi^{-1}(c)$$

PARAMETER MEANINGS

L	Daily loss, assumed Normal	μ_L	Mean of the daily losses	σ_L	Standard deviation of the daily losses
c	Confidence level	z_c	Standard Normal quantile at probability c	Φ^{-1}	Inverse standard Normal distribution function
ϕ	Standard Normal density function	VaR_c	Loss threshold at level c	ES_c	Mean loss beyond VaR under the Normal model

5.5 Monte Carlo: ordering and VaR

Simulated losses, ordered

$$L^{(1)} \geq L^{(2)} \geq \ldots \geq L^{(n)}$$

VaR position

$$j = \min\left\{ i \in \{1, \ldots, n\} : \frac{i}{n} > 1 - \alpha \right\}$$

Monte Carlo VaR

$$\widehat{\mathrm{VaR}}_{\alpha}(n) = L^{(j)}$$

PARAMETER MEANINGS

$L^{(i)}$	i-th simulated loss, sorted from largest to smallest	n	Number of simulated scenarios (100,000)	α	Confidence level; 1–α is the tail probability
i	Position in the ordered list	j	First ordered position that crosses the tail boundary	$\widehat{\mathrm{VaR}}_{\alpha}(n)$	Loss threshold estimated from the simulation

5.5 Monte Carlo: Expected Shortfall

Loss of a simulated profit change

$$L = -x$$

Monte Carlo ES

$$\widehat{\mathrm{ES}}_\alpha(n) = \frac{1}{1-\alpha}\left[\sum_{i=1}^{j-1}\frac{L^{(i)}}{n} + \left(1 - \frac{j-1}{(1-\alpha)\,n}\right)L^{(j)}\right]$$

PARAMETER MEANINGS

x	Simulated profit change
$L^{(i)}$	i-th simulated loss after sorting
j	First position crossing the tail boundary
L	Loss of the scenario
n	Number of simulated scenarios
$1 - \frac{j-1}{(1-\alpha)n}$	Weight of the j-th loss; zero if (1–α)n is an integer
$\widehat{\mathrm{ES}}_\alpha(n)$	Estimated average loss in the tail
α	Confidence level

5.6 Stress testing

Stressed sales

$$\text{Stressed Sales} = \text{Base Sales} \times (1 + \text{Sales Shock})$$

Stressed cost

$$\text{Stressed Cost} = \text{Base Cost} \times (1 + \text{Cost Shock})$$

Stressed profit

$$\text{Stressed Profit} = \text{Stressed Sales} - \text{Stressed Cost} - \text{Other Expenses}$$

Profit change

$$\text{Profit Change} = \text{Stressed Profit} - \text{Base Profit}$$

Profit change in %

$$\text{Profit Change \%} = \frac{\text{Stressed Profit} - \text{Base Profit}}{\text{Base Profit}} \times 100\%$$

PARAMETER MEANINGS

Base Sales	Observed annual sales	Base Cost	Observed annual cost	Base Profit	Observed annual profit
Sales Shock	Change in sales (−0.10 = 10% fall)	Cost Shock	Change in cost (0.10 = 10% rise)	Other Expenses	Costs not in the data; set to zero
Stressed Sales	Sales after the sales shock	Stressed Cost	Cost after the cost shock	Stressed Profit	Profit under the scenario

5.6 Stress rule of thumb and break-even

Rule of thumb

$$\text{Profit Change} = \text{Base Sales} \times \text{Sales Shock} - \text{Base Cost} \times \text{Cost Shock}$$

Sales decline that cancels all profit

$$\text{Sales Decline} = 1 - \frac{\text{Stressed Cost}}{\text{Base Sales}}$$

PARAMETER MEANINGS

Base Sales	Observed annual sales	Sales Shock	Scenario change in sales	Base Cost	Observed annual cost
Cost Shock	Scenario change in cost	Stressed Cost	Cost after the cost shock	Sales Decline	Fall in sales that wipes out all profit

5.7 Concentration risk

Revenue share

$$s_i = \frac{\text{Revenue}_i}{\text{Total Revenue}}$$

Largest share

$$\text{Largest Share} = \max(s_1, s_2, \ldots, s_n) \times 100\%$$

Top 5 share

$$\text{Top 5 Share} = \sum_{i=1}^{5} s_{(i)} \times 100\%$$

Herfindahl-Hirschman Index

$$\text{HHI} = \sum_{i=1}^{n} s_i^2 \times 10^4$$

PARAMETER MEANINGS

n	Number of entities (stores or categories)	i	Index identifying one entity	Revenue_i	Revenue of entity i
s_i	Share of total revenue of entity i (0 to 1)	$s_{(i)}$	Shares ordered from largest to smallest	Largest Share	Largest entity share, in %
Top 5 Share	Sum of the five largest shares, in %	HHI	Sum of squared shares times 10,000		

5.7 HHI floor and normalised HHI

Minimum HHI for n equal entities

$$\mathrm{HHI}_{\min} = \frac{10000}{n}$$

Normalised HHI

$$\mathrm{HHI}_{\mathrm{norm}} = \frac{\mathrm{HHI}/10000 - 1/n}{1 - 1/n}$$

PARAMETER MEANINGS

- n — Number of entities compared
- HHI — Observed index, from near 0 to 10,000
- $\mathrm{HHI}_{\min}$ — Lowest possible HHI with n equal entities
- $\mathrm{HHI}_{\mathrm{norm}}$ — HHI rescaled to the range 0 to 1

5.8 Average discount and margin rates

Average discount rate

$$\bar{D} = \frac{1}{n} \sum_{i=1}^{n} D_i$$

Average margin rate

$$\bar{M} = \frac{1}{n} \sum_{i=1}^{n} M_i$$

PARAMETER MEANINGS

i	Index identifying one transaction	D_i	Discount rate of transaction i	M_i	Margin rate of transaction i
n	Number of transactions included	$\bar{D}$	Arithmetic average of the discount rates	$\bar{M}$	Arithmetic average of the margin rates

5.8 Correlation between discount and margin

Pearson correlation

$$r = \frac{\sum_{i=1}^{n}(D_i - \bar{D})(M_i - \bar{M})}{\sqrt{\sum_{i=1}^{n}(D_i - \bar{D})^2 \sum_{i=1}^{n}(M_i - \bar{M})^2}}$$

t statistic

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}, \qquad df = n - 2$$

PARAMETER MEANINGS

D_i	Discount rate of transaction i	M_i	Margin rate of transaction i	n	Number of paired transactions
$\bar{D}$	Sample average discount rate	$\bar{M}$	Sample average margin rate	r	Pearson correlation, from –1 to 1
t	Tests whether r differs from zero	df	Degrees of freedom of the t statistic		

5.9 Demand volatility

NUMERATOR

$$\sigma$$

Standard deviation of the category's monthly demand

÷

DENOMINATOR

$$\mu$$

Mean monthly demand of the same category

=

RESULT

$$CV$$

Coefficient of variation: no unit, or in % after ×100

COEFFICIENT OF VARIATION

$$CV = \frac{\sigma}{\mu}$$

IN PERCENT

$$CV(\%) = \frac{\sigma}{\mu} \times 100$$

5.10 Entropy weights of the risk indicators

Min-max normalisation

$$z_{sj} = \frac{x_{sj} - \min_s(x_{sj})}{\max_s(x_{sj}) - \min_s(x_{sj})}$$

Share of each store

$$p_{sj} = \frac{z_{sj}}{\sum_{s=1}^{n} z_{sj}}$$

Entropy of indicator j

$$e_j = -\frac{1}{\ln(n)} \sum_{s=1}^{n} p_{sj} \ln(p_{sj})$$

Weight of indicator j

$$w_j = \frac{1 - e_j}{\sum_{k=1}^{5} (1 - e_k)}, \qquad \sum_{j=1}^{5} w_j = 1$$

PARAMETER MEANINGS

s	Index identifying a store	j	One of the five risk indicators	n	Number of stores
x_{sj}	Raw value of indicator j for store s	z_{sj}	Min-max normalised value (0 to 1)	p_{sj}	Share of store s in indicator j
e_j	Entropy of indicator j (ln = natural log)	w_j	Data-derived weight; the five weights sum to 1	k	Summation index over the indicators

5.10 Composite store risk score

Indicator score

$$S_{sj} = 100 \times PR_{sj} \qquad (100 - 100 \times PR_{sj} \text{ for margin risk})$$

Composite risk score

$$CRS_s = \sum_{j=1}^{5} w_j \, S_{sj}$$

Expanded form

$$CRS_s = w_{RV} S_{s,\,RV} + w_{MR} S_{s,\,MR} + w_{DR} S_{s,\,DR} + w_{CR} S_{s,\,CR} + w_{DV} S_{s,\,DV}$$

PARAMETER MEANINGS

PR_{sj}	Percentile rank of store s on indicator j	S_{sj}	0 to 100 risk score; reversed for margin	w_j	Entropy weight of indicator j
CRS_s	Composite risk score of store s (0 to 100)	RV	Revenue volatility	MR	Margin risk
DR	Discount risk	CR	Concentration risk	DV	Demand volatility

DEMO

Risk model walkthrough

DASHBOARD, FROM TRANSACTION DATA TO MANAGEMENT REPORTING

1

Data

Sales.csv: 2,000,000 lines from 21 stores

2

Quality control

Duplicates flagged, 1,500 invalid rows excluded

3

Risk drivers

Cost, profit, margin, discount and daily loss

4

Risk measurement

Historical, Parametric and Monte Carlo VaR and ES

5

Stress testing

Sales declines and cost increases up to 20%

6

Risk aggregation

Entropy-weighted risk score for each store

7

Management reporting

Risk register and recommendations

Thank You

Questions and discussion are welcome